

## Use Case “CUD Product”

### I. Use case “Create product”

#### 1. Use case code

UC004

#### 2. Brief Description

This use case describes the interaction between Product Manager and AIMS software when Product Manager wishes to create a new product in the system

#### 3. Actors

##### 3.1. Name of Actor 1

Product Manager

#### 4. Preconditions

- Product Manager has logged in to access the features corresponding to their role

#### 5. Basic Flow of Events

1. Product Manager requests to add a new product.
2. AIMS software displays the product information input form.
3. Product Manager enters the common product information, including title, category, general description, physical attributes, weight, barcode, original value, and current price.
4. Product Manager selects the product type: Book, Newspaper, CD, or DVD.
5. AIMS software displays the corresponding type-specific fields
6. Product Manager enters the required type-specific information and any optional additional information
7. Product Manager submits the product information.
8. AIMS software validates all entered data, including required fields, business rules, value ranges, and data format.
9. AIMS software saves the new product in the database.
10. AIMS software stores the history of product creation.
11. AIMS software displays a successful creation notification.

#### 6. Alternative flows

*Table 1 - Alternative flows of events for UC Create product*

| No | Location  | Condition                                                              | Action                                                                                                                 | Resume location                                                            |
|----|-----------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 1. | At Step 8 | If the input does not comply with the rules or the format is incorrect | - AIMS software notifies the Product Manager which data fields are invalid or missing, such as blank mandatory fields, | Resumes at step 3 (for common fields) or step 6 (for type-specific fields) |

|    |                                 |                                                   |                                                                                                                                                               |               |
|----|---------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|    |                                 |                                                   | invalid date format,<br>invalid dimensions or<br>weight, or current price<br>outside the allowed range.<br><br>- Product Manager<br>corrects the information. |               |
| 2. | At any step<br>before<br>Step 9 | If Product<br>Manager<br>cancels the<br>operation | AIMS software does not<br>create the product and<br>discards the unsaved input.                                                                               | Use case ends |

## 7. Input data

*Table A - Input data of product information*

| No | Data fields                | Description                                                    | Mandatory | Valid condition                                                  | Example                     |
|----|----------------------------|----------------------------------------------------------------|-----------|------------------------------------------------------------------|-----------------------------|
| 1. | Title                      | Name of the product                                            | Yes       | Not blank                                                        | Doraemon Vol. 1             |
| 2. | Category /<br>Product type | Product classification                                         | Yes       | Must be one of:<br>Book, Newspaper,<br>CD, DVD                   | Book                        |
| 3. | General description        | General description of the product, such as condition or color | Yes       | Not blank                                                        | New, yellow cover           |
| 4. | Physical attributes        | Height, width, length, and weight                              | Yes       | All values are positive numbers                                  | 20 cm x 10 cm x 3 cm; 200 g |
| 5. | Barcode                    | Product barcode                                                | Yes       | Not blank                                                        | 8934567                     |
| 6. | Original value             | Original value before VAT                                      | Yes       | Positive number; excludes 10% VAT                                | 100000 VND                  |
| 7. | Current price              | Selling price before VAT                                       | Yes       | Must be between 30% and 150% of original value; excludes 10% VAT | 90000 VND                   |

|     |                              |                                                               |                         |                                                                  |                                                                                          |
|-----|------------------------------|---------------------------------------------------------------|-------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 8.  | Book details                 | Author(s), cover type, publisher, publication date            | Yes if type = Book      | Required fields not blank                                        | Author: Smith; Cover type: Paperback; Publisher: Bach Khoa; Publication date: 15/03/2026 |
| 9.  | Book additional details      | Number of pages, language, genre                              | No                      | Optional                                                         | 250 pages; Vietnamese; Comic                                                             |
| 10. | Newspaper details            | Editor-in-chief, publisher, publication date                  | Yes if type = Newspaper | Required fields not blank                                        | Editor-in-chief: Nguyen Van A; Publisher: Thanh Nien; Publication date: 15/03/2026       |
| 11. | Newspaper additional details | Issue number, publication frequency, ISSN, language, sections | No                      | Optional                                                         | Weekly; ISSN 1234-5678                                                                   |
| 12. | CD details                   | Artists, record label, track list, genre                      | Yes if type = CD        | Required fields not blank; each track must have title and length | Artist: MCK; Record label: RPT; Track: 99%, 3m20s; Genre: Rap                            |
| 13. | CD additional details        | Release date                                                  | No                      | Optional                                                         | 20/05/2025                                                                               |
| 14. | DVD details                  | Disc type, director, runtime, studio, language, subtitles     | Yes if type = DVD       | Required fields not blank                                        | Disc type: Blu-ray; Director: Nguyen A; Runtime: 120 min                                 |
| 15. | DVD additional details       | Release date, genre                                           | No                      | Optional                                                         | 15/03/2024; Action                                                                       |

## 8. Output data

*Table B - Output data of product creation*

| No | Data fields                 | Description                                                                  | Display format     | Example                                  |
|----|-----------------------------|------------------------------------------------------------------------------|--------------------|------------------------------------------|
| 1  | Success confirmation        | Notification that the product was created successfully                       | Text               | "Product has been created successfully." |
| 2  | Invalid data notification   | Notification indicating missing or invalid fields or violated business rules | Text               | "Publication date format is incorrect."  |
| 3  | Created product information | Saved product information for confirmation                                   | Text / detail view | Title, type, barcode, current price      |

## 9. Postconditions

- A new product is successfully created and saved in the system.
- The history of product creation is stored and can later be queried by the Product Manager.
- If the operation is cancelled, no product is created and the system state remains unchanged.

## **II. Use case “Update product”**

### **1. Use case code**

UC005

### **2. Brief Description**

This use case describes the interaction between Product Manager and AIMS Software when the Product Manager wishes to update the information of an existing product.

### **3. Actors**

#### **3.1. Name of Actor 1**

Product Manager

### **4. Preconditions**

- Product Manager has logged in successfully
- The selected product already exists in the system

### **5. Basic Flow of Events**

1. Product Manager requests to update a product
2. AIMS software displays the list of products
3. Product Manager selects one product to update
4. AIMS software displays the current product information
5. Product Manager updates the product information
6. Product Manager submits the updated information
7. AIMS software validates the updated data, including required fields, business rules, value ranges, and format
8. AIMS software updates the product information in the database.
9. AIMS software stores the history of product update.
10. AIMS software displays a successful update notification.

### **6. Alternative flows**

***Table 2 - Alternative flows of events for UC Update Product***

| No | Location  | Condition                                                               | Action                                                                                 | Resume location  |
|----|-----------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------|------------------|
| 1  | At Step 3 | If the Product Manager attempts to edit more than one product at a time | AIMS software rejects the action and allows updating only one product at a time.       | Resume at Step 2 |
| 3. | At Step 7 | Updated information is invalid, incomplete, or wrongly formatted        | AIMS software notifies the Product Manager about the invalid fields or violated rules. | Resume at Step 5 |

|   |                           |                                       |                                            |               |
|---|---------------------------|---------------------------------------|--------------------------------------------|---------------|
| 3 | At any step before Step 8 | Product Manager cancels the operation | AIMS software does not update the product. | Use case ends |
|---|---------------------------|---------------------------------------|--------------------------------------------|---------------|

## 7. Input data

*Table C - Input data of product update*

| No | Data fields             | Description                                        | Mandatory          | Valid condition                                          | Example                                 |
|----|-------------------------|----------------------------------------------------|--------------------|----------------------------------------------------------|-----------------------------------------|
| 1  | Product ID              | ID of selected product                             | Yes                | Must exist in the system                                 | P001                                    |
| 2  | Title                   | Name of the product                                | Yes                | Not blank                                                | Doraemon Vol. 1                         |
| 3  | Category / Product type | Product classification                             | Yes                | Must be Book, Newspaper, CD, or DVD                      | Book                                    |
| 4  | General description     | General description of the product                 | Yes                | Not blank                                                | New edition                             |
| 5  | Physical attributes     | Height, width, length, and weight                  | Yes                | Positive numbers                                         | 20 cm x 10 cm x 3 cm; 220 g             |
| 6  | Barcode                 | Product barcode                                    | Yes                | Not blank                                                | 8934567                                 |
| 7  | Original value          | Original value before VAT                          | Yes                | Positive number                                          | 100000 VND                              |
| 8  | Current price           | Selling price before VAT                           | Yes                | Between 30% and 150% of original value; excludes 10% VAT | 95000 VND                               |
| 9  | Book details            | Author(s), cover type, publisher, publication date | Yes if type = Book | Required fields not blank                                | Smith; Hardcover; Bach Khoa; 15/03/2026 |

|    |                              |                                                               |                         |                                                            |                                      |
|----|------------------------------|---------------------------------------------------------------|-------------------------|------------------------------------------------------------|--------------------------------------|
| 10 | Book additional details      | Number of pages, language, genre                              | No                      | Optional                                                   | 250; English; Comic                  |
| 11 | Newspaper details            | Editor-in-chief , publisher, publication date                 | Yes if type = Newspaper | Required fields not blank                                  | Nguyen Van A; Thanh Nien; 15/03/2026 |
| 12 | Newspaper additional details | Issue number, publication frequency, ISSN, language, sections | No                      | Optional                                                   | No. 25; Weekly                       |
| 13 | CD details                   | Artists, record label, track list, genre                      | Yes if type = CD        | Required fields not blank; each track has title and length | MCK; RPT; Track 1: 99% - 3m20s       |
| 14 | CD additional details        | Release date                                                  | No                      | Optional                                                   | 20/05/2025                           |
| 15 | DVD details                  | Disc type, director, runtime, studio, language, subtitles     | Yes if type = DVD       | Required fields not blank                                  | Blu-ray; Nguyen A; 120 min           |
| 16 | DVD additional details       | Release date, genre                                           | No                      | Optional                                                   | 15/03/2024; Action                   |

## 8. Output data

*Table D - Output data of product update*

| No | Data fields          | Description                                            | Display format | Example                                  |
|----|----------------------|--------------------------------------------------------|----------------|------------------------------------------|
| 1  | Success confirmation | Notification that the product was updated successfully | Text           | "Product has been updated successfully." |

|   |                             |                                                                        |                        |                                                                 |
|---|-----------------------------|------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------|
| 2 | Invalid data notification   | Notification indicating invalid fields or violated business rules      | Text                   | "Current price must be between 30% and 150% of original value." |
| 3 | Updated product information | Updated product information for confirmation                           | Text / detail view     | Updated title, price, publication date                          |
| 4 | Update history record       | Confirmation that the update action has been logged in product history | Text / internal record | "Update history stored."                                        |

## 9. Postconditions

- Product information is successfully updated in the system
- The history of product editing is stored and can be queried by the Product Manager
- If the operation is cancelled, no product information is changed



### III. Use case “Delete product”

#### 1. Use case code

UC006

#### 2. Brief Description

This use case describes the interaction between Product Manager and AIMS software when the Product Manager wishes(es) to delete products from the system.

#### 3. Actors

##### 3.1 Name of Actor 1

Product Manager

#### 4. Preconditions

- Product Manager has logged in successfully
- The selected product already exists in the system

#### 5. Basic Flow of Events

1. Product Manager requests to delete product(s).
2. AIMS Software displays the product list.
3. Product Manager selects one or more products to delete.
4. AIMS Software checks that the number of selected products does not exceed 10 products per request.
5. AIMS Software checks that the Product Manager has not exceeded the limit of 20 deleted products per day.
6. AIMS Software checks the stock quantity of each selected product.
7. AIMS Software deletes all selected products whose stock quantity is equal to 0.
8. AIMS Software stores the deletion history.
9. AIMS Software notifies the Product Manager of the deletion result

#### 6. Alternative flows

*Table 3 - Alternative flows of events for UC Delete Product*

| No | Location  | Condition                                                           | Action                                                                                | Resume location   |
|----|-----------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|
| 1  | At Step 4 | If the Product Manager selects more than 10 products in one request | AIMS Software notifies the Product Manager that the request exceeds the allowed limit | Resumes at Step 3 |

|   |                           |                                                                                                                 |                                                                                                                                                                                                                                                                                       |                  |
|---|---------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 2 | At Step 5                 | If deleting the selected products causes the Product Manager to exceed the limit of 20 deleted products per day | AIMS Software notifies the Product Manager that the daily deletion limit has been exceeded                                                                                                                                                                                            | Use case ends    |
| 3 | At Step 6                 | If one or more selected products have stock quantity greater than 0                                             | <ul style="list-style-type: none"> <li>- AIMS software changes the status of those products to "deactivated" meaning it will no longer be available for sale</li> <li>- AIMS software stores this change in product history and notifies the Product Manager of the result</li> </ul> | Resume at Step 8 |
| 4 | At any step before Step 7 | If the Product Manager cancels the delete operation                                                             | AIMS Software does not delete or deactivate any product                                                                                                                                                                                                                               | Use case ends    |

## 7. Input data

**Table E - Input data of product deletion**

| No | Data fields          | Description                                                                       | Mandatory | Valid condition                                   | Example      |
|----|----------------------|-----------------------------------------------------------------------------------|-----------|---------------------------------------------------|--------------|
| 1  | Selected products    | List of products chosen for deletion                                              | Yes       | Maximum 10 products per request                   | [P001, P002] |
| 2  | Daily deletion count | Number of products already deleted on the current day by the same Product Manager | Yes       | Total deleted products per day must not exceed 20 | 12           |
| 3  | Stock quantity       | Current stock quantity of                                                         | Yes       | Product can only be deleted if stock = 0          | 0            |

|  |  |                       |  |  |  |
|--|--|-----------------------|--|--|--|
|  |  | each selected product |  |  |  |
|--|--|-----------------------|--|--|--|

  

## 8. Output data

*Table F - Output data of product deletion*

| No | Data fields              | Description                                                                | Display format         | Example                                               |
|----|--------------------------|----------------------------------------------------------------------------|------------------------|-------------------------------------------------------|
| 1  | Invalid notification     | Notification when the delete operation violates business rules             | Text                   | "Cannot delete more than 10 products in one request." |
| 2  | Daily limit notification | Notification when daily deletion limit is exceeded                         | Text                   | "Cannot delete more than 20 products per day."        |
| 3  | Product status           | Status shown for products that cannot be deleted because stock is not zero | Text                   | "Deactivated"                                         |
| 4  | Deletion result          | Result summary of the operation                                            | Text                   | "3 products deleted, 2 products deactivated."         |
| 5  | Deletion history record  | Confirmation that deletion/deactivation actions were logged                | Text / internal record | "Deletion history stored."                            |

  

## 9. Postconditions

- Selected products are deleted or deactivated in the database
- The history of product deletion is stored and can be queried by the Product Manager
- If the operation is cancelled or rejected due to rule violations, the system state remains unchanged